

```

3672.663459] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3672.683513] usb usb6-port1: config error
3676.783452] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3676.783452] usb usb6-port1: config error
3680.743479] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3680.743436] usb usb6-port1: config error
3684.783453] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3684.783452] usb usb6-port1: config error
3688.823457] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3688.823420] usb usb6-port1: config error
3692.863458] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3692.867222] usb usb6-port1: config error
3696.906738] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3696.906794] usb usb6-port1: config error
3700.946740] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3700.946911] usb usb6-port1: config error
3704.986690] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3704.986747] usb usb6-port1: config error
3709.033459] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3709.033453] usb usb6-port1: config error
3713.073415] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3713.073469] usb usb6-port1: config error
3717.113477] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3717.113428] usb usb6-port1: config error
3721.153499] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3721.153455] usb usb6-port1: config error
3725.193417] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3725.193443] usb usb6-port1: config error
3729.233433] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3729.233488] usb usb6-port1: config error
3733.273499] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3733.273428] usb usb6-port1: config error
3737.313443] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3737.313498] usb usb6-port1: config error
3741.353448] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3741.353427] usb usb6-port1: config error
3745.393498] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3745.393452] usb usb6-port1: config error
3749.433455] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3749.433444] usb usb6-port1: config error
3753.473457] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3753.473413] usb usb6-port1: config error
3757.513462] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3757.513424] usb usb6-port1: config error
3761.553491] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3761.553446] usb usb6-port1: config error
3765.593450] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3765.593448] usb usb6-port1: config error
3769.633400] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3769.633454] usb usb6-port1: config error
3773.673484] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
3773.673440] usb usb6-port1: config error

```

现象:

在进入系统之后,如果USB3.0_A接口插升级线连接着PC,那么可以后台查出大量循环打印的usb错误信息并且下发lsusb命令时,可能会卡住几秒到十几秒

日志记录:

```

[ 3729.233433] usb usb6-port1: Cannot enable. Maybe the USB cable is bad?
[ 3729.233488] usb usb6-port1: config error

```

关联内核代码位置:

```
./drivers/usb/core/hub.c
```

相关函数:

hub_port_reset

port_event

hub_event

日志直接打印位于hub_port_reset函数中

```

3      |         delay = HUB_LONG_RESET_TIME;
2      |     }
1      |
3017   |     dev_err(&port_dev->dev, "Cannot enable. Maybe the USB cable is bad?\n");
1      |
2      | done:
3      |     if (status == 0) {
4      |         if (port_dev->quirks & USB_PORT_QUIRK_FAST_ENUM)

```

从日志打印处往回查,调用这个函数的角色是:port_event

```

f hub_set_initial_usb2_cpm_ 7
f hub_enable_device 8 /*
f get_bMaxPacketSize0 7 * Warm reset a USB3 protocol port if it's in
f hub_port_init 6 * SS.Inactive state.
f check_highspeed 5 */
f hub_power_remaining 4 if (hub_port_warm_reset_required(hub, port1, portstatus)) {
f descriptors_changed 3 dev_dbg(&port_dev->dev, "do warm reset\n");
f hub_port_connect 2 if (!udev || !(portstatus & USB_PORT_STAT_CONNECTION)
f hub_port_connect_change 1 && udev->state == USB_STATE_NOTATTACHED) {
f port_over_current_notify 5654 if (hub_port_reset(hub, port1, NULL,
f port_event 1 > HUB_BH_RESET_TIME, true) < 0)
f hub_event 2 hub_port_disable(hub, port1, 1);
f usb_hub_init 3 } else {
f usb_hub_cleanup 4 usb_unlock_port(port_dev);
f usb_reset_and_verify_dev 5 usb_lock_device(udev);

```

它会在内部尝试重启端口,并且里面固定了循环重建的尝试次数和每次尝试的间隔,在超时会进入hub_port_disable关闭这个端口,之后在走到函数末尾后退出port_event

但是由于线仍然插着,会触发新的事件重新调用hub_event,也是在这个函数里给根集线器上锁了(和lsusb同一把锁),也就是说在之前循环reset期间lsusb都会卡住等待窗口

lsusb 阻塞在等待hub_event函数内

```

> (u16) hub->change_bits[0],
> (u16) hub->event_bits[0]);

/* Lock the device, then check to see if we were
 * disconnected while waiting for the lock to succeed. */
usb_lock_device(hdev);
if (unlikely(hub->disconnected))
    goto out_hdev_lock;

/* If the hub has died, clean up after it */

```

这把锁,它会在经过hub_port_reset()中等待约4秒之后返回处理完自身状态之后释放,直到下一次事件触发

整个过程:

USB3 PHY检测到usb对端设备电气属性->xHCI硬件中断将hub_event加入工作队列->先给根集线器上锁再给端口上锁(影响lsusb的是根集线器锁)-> 进入port_event然后进入端口复位函数-> hub_port_reset重试耗尽->回到port_event并关闭这个端口->运行到函数底部返回hub_event函数,在hub_event末尾解锁根集线器->线还插着,物理接口再次触发事件回到起点

如果 lsusb 恰好在那个很短的释放窗口(hub_event解锁后到下次事件触发进入hub_event函数头部上锁之间)获得集线器锁,就会快速返回;如果没有抢到,就要再等一轮甚至多轮。